Tumour accumulation is not tumour-to-liver selectivity: a publication-aware polymer nanoparticle reanalysis

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Abstract

Selecting a polymer nanoparticle solely for tumour accumulation can ignore competing liver uptake. We quantify the relationship between these ranking objectives using paired whole-organ biodistribution measurements from the publicly archived Nano-Tumor Database. The primary cohort contains 142 source-labelled Polymeric formulations in 72 connected publication-reference groups, evaluated at 24 hours without temporal extrapolation. Analyses include publication-cluster bootstrap intervals, within-publication active/passive contrasts, fixed baseline prediction models and prespecified time/material sensitivity analyses. Median tumour and liver accumulation were 0.625%ID and 9.750%ID. Tumour accumulation and the tumour/liver ratio had Spearman correlation 0.561 (publication-bootstrap 95% interval 0.396 to 0.707); only 47.2% of the top-quartile formulations overlapped between objectives. The equal-weight within-publication active/passive log10-ratio contrast was 0.209 (0.091 to 0.325) across 15 overlapping publication groups. Additional objective-weight, publication-omission and capacity ablations preserve the primary cohort. Leave-one-group-out within-publication contrasts range from 0.183 to 0.244 log10 units. The observational contrast remains confounded, and organ selectivity is not a validated safety or efficacy endpoint.

Keywords: polymer nanoparticles; tumour biodistribution; liver uptake; active targeting; publication clustering.

1 Introduction

Nanoparticle delivery is a multiorgan distribution problem. Tumour accumulation may coexist with substantial clearance-organ uptake, and a material selected for one endpoint need not be preferred under another. Broad nanoparticle design and transport literature emphasizes biological barriers and model dependence (Mitchell et al., 2021; Wilhelm et al., 2016; Sindhwani et al., 2020).

Chen et al. (2023a) assembled an expanded Nano-Tumor Database and already analyzed tumour and major-organ distribution with multiple pharmacokinetic metrics and statistical models. The dataset, experimental curation and general organ-competition question were established in that work. Polymeric-micelle targeting has also been the subject of a 2026 mouse-model meta-analysis by Sousa-Oliveira and colleagues. These resources enable a polymer-specific comparison of paired accumulation objectives.

Polymer architecture, hydrodynamic size, surface properties and the biomolecular corona can all affect biodistribution. Cabral et al. (2011) studied size-dependent polymeric micelle accumulation, while Salvati et al. (2013) demonstrated a context in which a protein corona impaired targeting capability. These findings motivate caution, not a universal material rule or a general claim that ligands do not work.

We conduct an explicitly paired, same-unit, 24-hour polymer-specific ranking audit combined with connected-reference grouping and a separation of pooled and within-publication targeting associations. It asks: how concordant are tumour uptake and tumour/liver selectivity rankings; how does the active/passive association depend on the comparison unit; and can low-compute predictors transfer across source publications?

The scope is cancer biodistribution in mice, not brain targeting. No blood-brain barrier transport measurement or brain-tumour-specific efficacy dataset is validated here. The underlying data cannot support a claim that a new polymer crosses the barrier, treats cancer or is safe for humans.

A deployment question and a screening objective are separate design choices. The deployment question specifies which experimental context must be new at evaluation, whereas the objective specifies

which measured outcome a candidate should optimize. Both choices should be explicit before interpreting a favourable average score. Here we retain the original publication-disjoint benchmark and add exploratory diagnostics that probe information content, sample efficiency and decision sensitivity without replacing unfavourable primary results. These extensions were developed after inspecting the primary analysis and are not preregistered confirmatory tests.

2 Public data, pairing and polymer cohort

The primary source is the ACS Figshare supporting workbook nn3c04037_si_002.xlsx, DOI 10.1021/acsnano.3c04037.s002, associated with Chen et al. (2023a). Repository metadata specify CC BY-NC 4.0. This is a noncommercial secondary research analysis; commercial reuse needs separate license review. File size and source MD5 are checked during fetching and the analysis additionally records a SHA-256 digest.

Cohort quantity	Count
Source formulation groups with measurement rows	534
Source measurement rows retained	2345
All-material paired endpoints at 24 h	378
Primary Polymeric endpoints	142
Primary publication-reference groups	72

Table 1: Executed cohort flow. (Part 1 of 2.)

Cohort quantity	Count
Primary endpoints measured exactly at 24 h	142

Table 1: Executed cohort flow. (Part 2 of 2.)

The sheet has a two-row header identifying Biodistribution (%ID) and named organs. Only tumour and liver values from the same formulation and same source time enter the paired curve. Duplicate same-time pairs are averaged. The endpoint is interpolated only within that paired curve's observed range. No organ is imputed from an unrelated formulation. Positive values no greater than 100%ID are required; zeros are excluded without a pseudocount.

The primary material rule is exact source category 'Polymeric', not a text search selected for good outcomes. Hydrogel and Dendrimer categories enter a declared broader sensitivity cohort; other organic and hybrid materials are retained in source audits but are not silently relabelled as polymeric. Publication references such as 67/200 link both source references: their connected component is used for all resampling and prediction splits.

Analysis hierarchy. The formulation is the prediction unit, but the publication-reference group is the partitioning unit. Multiple formulations can share experimental protocols and reporting conventions, so counting them as independently sampled research settings would overstate the information available. We retain all eligible formulations within a selected group. Group sizes therefore vary across partitions and learning-curve subsets; nominal percentages of groups should not be read as exact percentages of formulations. No target-dependent exclusion is added to the primary cohort during the model comparisons.

3 Endpoints, estimands and statistical design

Let T and L be paired 24-hour whole-organ tumour and liver accumulation in percent injected dose. Define R=T/L and Z=log10(R). Both numerator and denominator use the same source unit; no %ID/g tissue concentration is substituted. A higher ratio can reflect more tumour uptake, less liver uptake or both. It is not necessarily better therapy, and low absolute delivery can coexist with a high ratio.

The primary ranking audit reports Spearman correlation between T and R, and overlap between their top ceil(n/4) formulation sets. Ties follow the stable source-row ordering in pandas nlargest. Publication-cluster bootstrap (2,000 replicates, seed 603) summarizes uncertainty in Spearman correlation. The top-quartile overlap is descriptive and has no claimed confidence interval.

Two active/passive estimands are distinguished. The pooled formulation-weighted contrast is mean(Z|Active)-mean(Z|Passive), with 2,000 publication-block bootstrap replicates. The within-publication estimand first computes that difference in each publication group containing both categories, then averages publication differences equally. A 5,000-replicate bootstrap resamples those differences. These are associations, not randomized or composition-matched ligand effects.

Fixed prediction baselines use hydrodynamic

diameter, zeta potential, shape, source material category, targeting category and source tumour-cell description. Only an explicitly labelled HD value is treated as hydrodynamic size; TEM-only measurements remain missing. Training-only imputation, scaling and one-hot encoding precede ridge (alpha=10) or 128 extra trees (minimum leaf=3, one worker). Unseen categories are ignored by the encoder.

Five seeds partition publication groups into approximately 60% training, 20% reserved/unused and 20% test, matching the research scaffold without using the reserved outcomes for tuning. MAE in Z, study-macro MAE and R^2 are reported. Time sensitivities at 6 and 48 hours and an expanded material cohort are prespecified descriptive analyses, not independent replication. No multiple-testing-adjusted discovery claim is made.

Exploratory extension design. Every additional configuration uses the saved five publication partitions. Ridge penalties of 1, 10 and 100 probe regularization; tree leaf sizes of 1, 3 and 8 probe local averaging; 32 versus 128 trees probes ensemble size. Feature-block removal is evaluated separately from changing model capacity. All configurations are reported rather than choosing a winner using held-out errors. Consequently this is a sensitivity grid, not a hyperparameter-selection procedure or a claim about the optimally achievable prediction error.

Paired comparisons. Within a seed, configurations share the same training and test formulations. Their MAE differences can therefore be compared on a common evaluation composition. Across seeds, however, publications are reused. The five scores are descriptive repeated-partition results, not independent replications. We report their full values and dispersion, and avoid significance tests that treat the five partitions as independently sampled laboratories.

$$z_i(\lambda) = \log_{10} T_i - \lambda \log_{10} L_i. \tag{1}$$

At lambda=0 the objective ranks tumour accumulation alone; at lambda=1 it ranks the tumour/liver ratio. Other weights are exploratory preferences for liver avoidance, not measured clinical utility. Ranking candidates under several weights exposes a choice that is otherwise hidden inside a scalar score. Both organ measurements remain in the same whole-organ %ID units throughout this comparison.

$$d_g = \overline{z}_{g,\mathrm{Active}} - \overline{z}_{g,\mathrm{Passive}},$$
$$\widehat{\Delta}_{\mathrm{within}} = \frac{1}{G_{\mathrm{both}}} \sum_{g=1}^{G_{\mathrm{both}}} d_g. \tag{2}$$

4 Tumour uptake and selectivity rank different formulations

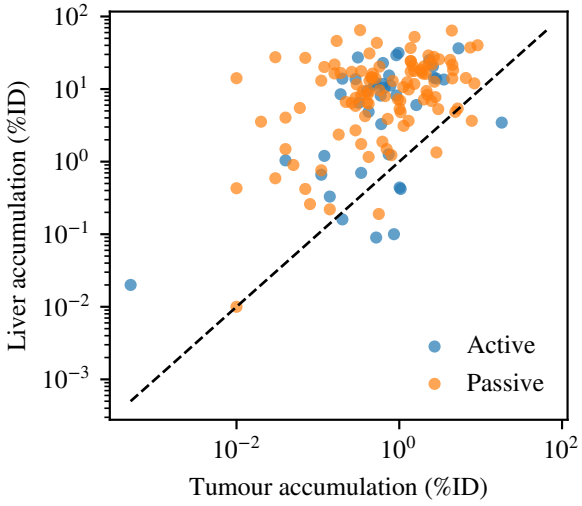


Figure 1: Paired 24-hour tumour and liver %ID for the primary polymer cohort. The diagonal is equal whole-organ accumulation, not a toxicity threshold.

Figure 1 shows the paired organ measurements. The median tumour/liver ratio was 0.0915. The median of ratios is not generally equal to the ratio of separate tumour and liver medians, because pairing matters. The 95% publication-bootstrap interval for the median log10 ratio was -1.227 to -0.898; this interval targets the literature-selected cohort summary, not a patient-level therapeutic index.

Rank correlation was 0.561, rather than perfect concordance. The two top-36 lists shared 17 formulations. Thus a large part of a top-quartile shortlist changes when the objective switches from accumulation to organ selectivity. This does not identify which formulation should be used clinically.

T appears in both ranking variables, so positive correlation is partly mathematical and is not evidence that tumour uptake causes favourable selectivity. Liver uptake, time, assay design, tumour model and formulation composition contribute further variation. Whole-organ ratios are useful as a transparent screening objective only when accompanied by absolute exposure and an explicit account of what they omit.

5 Targeting associations depend on the comparison unit

strategy	Formul. n	Pub. n	median tumor percent id
Active	39	25	0.6200
Passive	103	62	0.6700

Table 2: Primary cohort summaries by source targeting category. (Part 1 of 2.)

strategy	median liver percent id	median log10 ratio
Active	8.4800	-1.0296
Passive	10.6700	-1.0476

Table 2: Primary cohort summaries by source targeting category. (Part 2 of 2.)

Association estimand	log10 contrast	95% lower
Pooled formulation contrast	0.2346	-0.1116
Equal-weight publication contrast	0.2088	0.0914

Table 3: Active-minus-passive log10 selectivity associations. Percentile intervals are publication-aware, not causal. (Part 1 of 2.)

Association estimand	95% upper	Groups
Pooled formulation contrast	0.6054	72
Equal-weight publication contrast	0.3254	15

Table 3: Active-minus-passive log10 selectivity associations. Percentile intervals are publication-aware, not causal. (Part 2 of 2.)

The pooled contrast was 0.235 log10 units with a bootstrap interval spanning -0.112 to 0.605. That interval includes zero. The within-publication summary was 0.209 (0.091 to 0.325), but uses only publications reporting both categories and a different weighting scheme. These estimands answer different questions.

Exponentiating the within-publication contrast gives a descriptive 1.62-fold ratio of geometric selectivity (1.23 to 2.12). It must not be reported as a causal improvement, an efficacy fold-change, or a recommendation to add a particular ligand. Active and passive formulations can differ in chemistry, size, dose, surface treatment and assay conditions.

Restricting to publications containing both categories reduces some between-publication heterogeneity but does not ensure matched polymer architecture or randomized animal allocation. Publication selection, outcome-dependent reporting and the small number of overlapping groups remain important. No study-level animal counts or standard errors are sufficiently harmonized here for an animal-level meta-analysis.

The exact-24-hour restriction reproduces the primary cohort because every retained primary polymer endpoint is observed at 24 hours. Accordingly, that check verifies endpoint timing but is not an independent robustness dataset and must not be counted as a replication.

6 Prediction, ablations and sensitivity

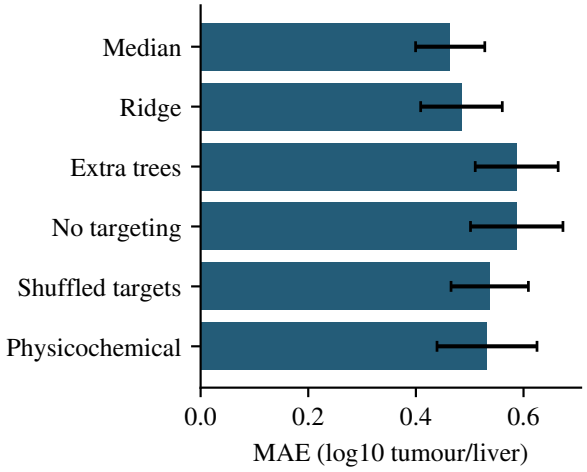


Figure 2: Publication-separated prediction of log10 tumour/liver ratio. Error bars denote SD across overlapping partitions.

Figure 2 compares the prediction baselines and controls. Primary extra-trees MAE was 0.587 log10 units, versus 0.464 for the training-median reference and 0.485 for ridge. Tree mean R^2 was -0.585. These results do not demonstrate a reliable prospective selectivity predictor. The complete ablation table retains no-targeting, physicochemical-only and target-permuted conditions rather than selecting the most favourable result.

hours / cohort	n	Pub. n
6 / polymeric	163	84
6 / polymer extended	187	99
24 / polymeric	142	72
24 / polymer extended	166	87
48 / polymeric	61	32
48 / polymer extended	77	43

Table 4: Time and material sensitivities. Changing time also changes the eligible cohort. (Part 1 of 2.)

hours / cohort	Rank correlation	Top-list overlap
6 / polymeric	0.5141	0.4634
6 / polymer extended	0.5478	0.5106
24 / polymeric	0.5610	0.4722
24 / polymer extended	0.6043	0.5238
48 / polymeric	0.4757	0.4375
48 / polymer extended	0.6064	0.5500

Table 4: Time and material sensitivities. Changing time also changes the eligible cohort. (Part 2 of 2.)

The formulation-random diagnostic allows the same source publication to recur across training and test. It is reported but is not a prospective validation scheme. Removing targeting from predictors tests a feature block, not whether targeting has a causal biological effect. Weak or improved ablation performance cannot resolve that question.

The 6- and 48-hour comparisons conflate temporal evolution with changing observation availability. The broader cohort adds hydrogels and dendrimers, not arbitrary polymer-coated inorganic hybrids. No time-dependent causal or universal material conclusion is inferred.

Model-capacity sensitivity. The extended grid holds the source cohort, partition manifests and scoring definitions fixed. A smaller ensemble changes Monte Carlo averaging but does not add process measurements. Changing leaf size changes the degree of local smoothing. A stronger ridge penalty changes shrinkage after training-only standardization. These interventions distinguish sensitivity to the learner from sensitivity to the available descriptors.

Setting	MAE	R2
Trees 128 leaf3	0.5873	-0.5850
Trees 128 leaf1	0.5911	-0.6877
Trees 128 leaf8	0.4927	-0.1216
Trees 32 leaf3	0.5925	-0.6138
Ridge alpha1	0.5262	-0.2679
Ridge alpha10	0.4847	-0.1328
Ridge alpha100	0.4661	-0.0636

Table 5: Capacity and regularization sensitivity on the original publication holdouts.

The tree leaf-size variants had mean MAEs of 0.591, 0.587, and 0.493 for leaf sizes 1, 3 and 8. The 32-tree result was 0.592. These are observations about the fixed grid, not evidence that the best observed configuration will retain its ranking on another dataset. The complete seed-level scores, including group-weighted MAE, are reported in Appendix M.

7 Selection objectives and multiorgan trade-offs

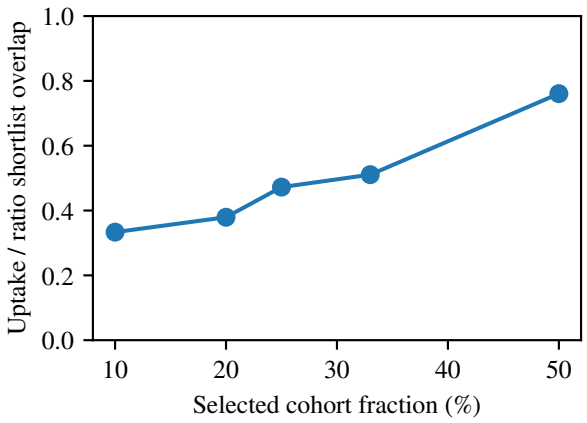


Figure 3: Uptake-versus-ratio shortlist overlap across selection fractions in the fixed primary cohort.

Figure 3 repeats the shortlist comparison at selection fractions of 10%, 20%, 25%, 33% and 50%, using the ceiling of the requested fraction times the cohort size. A change in the overlap at different fractions need not indicate a change in biology: the lists become longer and eventually must overlap more. Reporting several thresholds shows whether the original top-quartile result is specific to one shortlist length.

At the 10% and 50% fractions, uptake-versus-ratio shortlist overlaps were 33.3% and 76.1%, respectively. The paired high-tumour/low-liver Pareto frontier contains 6 formulations. A frontier point has no other observed formulation with

at least as much tumour accumulation and at most as much liver accumulation, with one strict improvement. Frontier membership does not establish safety or superiority under different experimental conditions.

liver weight	uptake overlap	ratio overlap	median tumor
0.0000	1.0000	0.4722	2.8500
0.2500	0.8889	0.5833	2.8500
0.5000	0.6944	0.7778	2.8100
1.0000	0.4722	1.0000	1.4600
1.5000	0.3056	0.8333	0.7700
2.0000	0.2222	0.7222	0.5800

Table 6: Liver-penalty sensitivity for top-quartile selection; organ medians remain %ID. (Part 1 of 2.)

liver weight	median liver
0.0000	15.2500
0.2500	13.7300
0.5000	7.0150
1.0000	3.2800
1.5000	1.3000
2.0000	1.0300

Table 6: Liver-penalty sensitivity for top-quartile selection; organ medians remain %ID. (Part 2 of 2.)

The weighted objectives and Pareto calculation use measurements from different publications. They describe the recorded dataset, not a head-to-head carrier trial. A candidate can look attractive because its assay time, dose or tumour model differs from another candidate. The frontier is therefore an illustration of the selection objective, not a recommendation to synthesize or administer a listed material.

8 Publication influence, denominator tails and confounding

We omit each connected publication group in turn and recompute the pooled contrast, equal-weight within-publication contrast and rank correlation. This identifies sensitivity to individual source groups without calling the omitted datasets independent validation cohorts. These omission results are not bootstrap confidence intervals.

The within-publication contrast ranged from 0.183 to 0.244 log10 units under group omission. Complete-physicochemical-case and minimum-liver subsets are evaluated separately below. A liver floor is a numerical sensitivity analysis, not a detection limit supplied by the source assay. Excluding small denominators changes the target population.

cohort	n	groups	within contrast
Primary	142	72	0.2088
Complete size and zeta	119	57	0.1465
Liver at least 0.01	142	72	0.2088
Liver at least 0.1	139	70	0.2088
Liver at least 1	124	61	0.2026

Table 7: Cohort and denominator sensitivities of the observational association.

Hydrodynamic size and zeta-potential balance are reported in Appendix S, including standardized mean differences and within-group numerical contrasts. Publication restriction does not guarantee composition-matched carriers: active and passive labels can still differ in architecture, surface treatment, dose and assay conditions. These diagnostics expose possible confounding rather than estimating a causal ligand effect or correcting unmeasured covariates.

A shuffled-liver diagnostic produced median rank correlation 0.704 between tumour uptake and the reconstructed ratio. This experiment retains the shared tumour numerator while breaking the observed tumour/liver pairing. It is a mathematical-coupling diagnostic, not a valid biological null model or a publication-adjusted permutation test.

9 Data efficiency and descriptor support

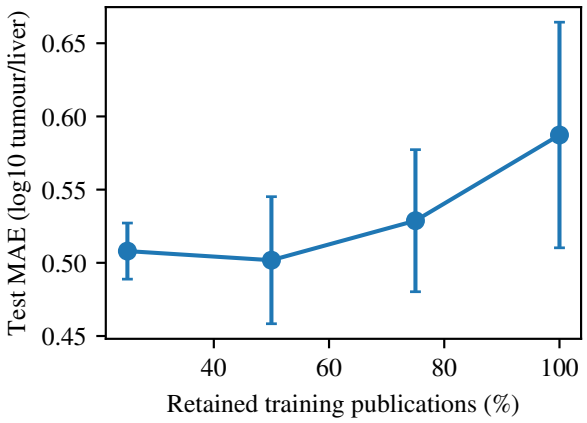


Figure 4: Nested publication-group learning sensitivity. Error bars are SD across overlapping outer partitions.

Figure 4 retains nested subsets of the original training publications at four fractions while keeping each seed's test set fixed. The reserved 20% partition is not added to training or used to select

a preferred configuration. Complete publication groups, rather than individual formulations, enter each nested training subset.

Mean test MAE was 0.508 at 25% training-group retention and 0.587 with all original training groups. This curve measures sensitivity to the available literature mixture, not the benefit of acquiring an equal number of new matched experiments. A larger training set can introduce new tumour labels and measurement conventions as well as additional examples of familiar carriers.

Train fraction	mae	Groups
0.2500	0.5080	11.0000
0.5000	0.5018	21.0000
0.7500	0.5288	32.0000
1.0000	0.5873	42.0000

Table 8: Full-data-efficiency summary on the fixed publication holdouts. (Part 1 of 2.)

Train fraction	Formulations
0.2500	22.2000
0.5000	43.2000
0.7500	65.6000
1.0000	83.4000

Table 8: Full-data-efficiency summary on the fixed publication holdouts. (Part 2 of 2.)

Numeric support checks flag size or zeta values outside the training range. Categorical checks flag unseen targeting, shape, core or tumour-cell labels. An unseen category is encoded without adding a new outcome-informed column. Missing measurements are tracked separately. These checks audit the recorded representation; they do not measure support for polymer chemistry that the dataset does not encode. Appendix O reports every feature and partition.

10 Discussion and data availability

The data support separating tumour accumulation from tumour/liver selectivity when reporting polymer nanoparticle screens. They also show that a pooled targeting contrast and a within-publication contrast need not have the same interpretation. These are modest methodological observations built on an existing database, not evidence for a new polymer mechanism or treatment.

The resource covers published mouse experiments with heterogeneous tumour models, doses, sampling times and assays. A connected reference group is not a verified independent laboratory. %ID is whole-organ accumulation and does not measure free drug, cellular internalization, tumour subregions, organ toxicity or pharmacodynamic benefit. Liver is only one competing organ. Excluding zero or missing paired measurements may create selection bias.

Independent evaluation should use composition-matched active/passive comparisons with drug exposure, tumour response and toxicity endpoints. Brain targeting would require separate evidence of barrier transport and relevant tissue distribution. None of those experiments has been performed here.

Availability and compute. Attributed source metadata, hashes, analysis cohorts, reference-group manifests, predictions, ablations and bootstrap summaries accompany the code. Instrumented analysis took 275.5 seconds with sampled process RSS 226.2 MB and one worker. Timing excludes setup, downloads, pre-context imports and manuscript generation. Dataset noncommercial restrictions remain in force.

This secondary public-data analysis generated no new animal or human observations.

The additional analyses used 87.4 instrumented seconds and sampled peak process RSS 184.9 MB with one worker. The primary cohort and original results remain unchanged; the extension independently reproduces the original publication-split tree scores before interpreting additional comparisons. Setup and manuscript compilation are excluded.

The additional analyses separate three potential sources of instability: the screening objective, the observational cohort and the fitted predictor. Changing a liver penalty changes what is selected; omitting a publication changes the literature mixture; and changing model capacity changes the mapping from the recorded descriptors to the same target. None of these interventions substitutes for measuring free-drug exposure or therapeutic response. A future matched study should specify its multiorgan objective before using a predictor to choose which carriers receive further assays.

The numerator and denominator should also be interpreted separately. A higher tumour/liver ratio can result from greater tumour accumulation, lower liver accumulation, or a decrease in both organs with a larger relative decrease in liver. These scenarios are not interchangeable for a drug that needs an absolute intratumoural exposure threshold. Whole-organ %ID does not distinguish an intact

carrier from a released label or quantify unbound drug at its molecular target. The ratio is therefore most defensible as one component of an explicitly multivariate screening report that retains absolute organ values.

Likewise, lower liver accumulation is not synonymous with lower toxicity. A material can redistribute to spleen, lung, kidney or circulating compartments, and organ injury can depend on local concentration, degradation products and exposure duration. The source endpoint is a single 24-hour whole-organ observation, not a time-integrated concentration curve. The proposed objective does not account for delayed clearance or repeated dosing. A screening programme could report paired tumour and liver values alongside a wider organ panel, then evaluate toxicity as a separate measured outcome rather than inferring it from the selectivity score.

Covariate adjustment would require a biological estimand in addition to a statistical model. Size and surface charge can be consequences of ligand conjugation as well as differences between independently designed carriers. Treating them automatically as confounders could remove part of a formulation's total effect, while omitting them can leave composition imbalance. The current balance table does not resolve that causal structure. A matched comparison should first specify whether the question concerns adding a ligand to a fixed carrier or comparing complete carrier designs that differ in several properties.

The recorded tumour-cell descriptor was unseen in training for a mean 53.5% of held-out formulations. This is a property of the publication-separated representation, not an estimate of biological novelty. Coarse source labels can conceal substantial heterogeneity, whereas unfamiliar labels can sometimes describe related biological settings. The result motivates richer, harmonized tumour-model metadata rather than allowing source identity to leak into training. Negative predictive R^2 should be interpreted as weak performance of this recorded-feature benchmark, not evidence that polymer design has no effect on distribution.

A discriminating follow-up would compare composition-matched carriers across laboratories while varying ligand presence and density independently of particle size and drug loading. Assays would distinguish carrier-associated signal from free drug, measure several common sampling times, and record tumour response and organ-specific injury without substituting accumulation for either endpoint. The screening objective, useful exposure threshold and acceptable competing-organ exposure would be specified before evaluating prediction. Such a design could test whether a selectivity-aware shortlist changes experimental decisions, whereas the present audit establishes only how the recorded screening choices differ.

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B Data dictionary and unit contract

Variable	Unit / representation	Interpretation
id	Source dataset number	Formulation/ time-course identifier
study	Connected reference label	Grouping only; not a predictor
tumor / liver	% injected dose	Paired whole-organ accumulation
hours	h	Source time; primary endpoint=24
log ratio	log10(T/L)	Prediction target and contrast scale
size nm	nm	Explicit source HD only; TEM-only missing

Table S1: Analysis variables and units. (Part 1 of 2.)

Variable	Unit / representation	Interpretation
zeta mv	mV	Strict numeric source zeta potential
strategy	Active / Passive	Source label, not a causal assignment
shape / core	Source category	One-hot encoded in training
tumor cell	Source text category	Heterogeneous model description
exact time	Boolean	Paired 24-h source observation present

Table S1: Analysis variables and units. (Part 2 of 2.)

A same-unit ratio is essential. The original paper discusses both %ID and %ID/g metrics, but this workbook's paired sheet explicitly labels the selected biodistribution block %ID. No concentration-unit value is substituted from a separate model table. A whole-organ ratio cannot be interpreted as a tissue-normalized concentration ratio.

The parser checks organ headers before using fixed columns. It accepts numeric source references or explicit slash-linked numeric references and rejects unresolved labels. Missing or ambiguous source fields must be reported, not repaired by guessing a number.

C Cohort construction and exclusion audit

First-failure reason at 24 h	Formulations
no paired bracket	151
nonpositive or above 100 percent ID	5

Table S2: Exclusions before the primary material filter.

Predictor	Missing in primary cohort
size nm	11
zeta mv	12

Table S3: Missing numerical predictors.

The source long table uses blank repeated metadata cells within a formulation time course. Dataset number is forward filled, and each metadata column takes its first available value within that dataset group. Metadata are not filled across distinct formulation groups. Only rows with a recognized ma-

terial type and a numeric measurement time enter the observation audit.

The endpoint builder retains only times containing both tumour and liver observations. This ensures interpolation uses paired source support rather than independently interpolating across incompatible organ observation windows. At 24 hours all retained primary polymer formulations have an exact paired observation.

Nonpositive and above-100 %ID endpoints are excluded with explicit reasons. No arbitrary epsilon is added to manufacture a finite logarithm. The exclusions file identifies source dataset numbers. The resulting selection can bias population summaries if low accumulation is preferentially missing or reported as zero.

D Material categories and primary endpoint distribution

Core category	24-h pairs
Polymeric	142
Gold	54
Liposome	34
Silica	26
Hydrogel	15
Other-Graphene	9
Dendrimer	9
Iron Oxide	8
Other-MWCNT	5
Other-Anticancer drug	4
Other-SLN	4

Table S4: Most frequent all-material endpoint categories; remaining source labels stay in the machine-readable audit. (Part 1 of 2.)

Core category	24-h pairs
Copper	4
Gadolinium	4
Manganese	4

Table S4: Most frequent all-material endpoint categories; remaining source labels stay in the machine-readable audit. (Part 2 of 2.)

index	tumor	liver	log ratio
count	142.0000	142.0000	142.0000
mean	1.4668	12.8807	-1.0347
std	2.2435	12.3082	0.7002
min	0.0005	0.0100	-3.1498
25%	0.2900	3.7075	-1.4374
50%	0.6250	9.7500	-1.0386
75%	1.8725	17.6000	-0.6474
max	18.1400	64.8600	0.9345

Table S5: Primary polymer endpoint distributions.

The primary core category is Polymeric exactly as supplied by the source. This avoids reclassifying polymer-coated inorganic or multi-component particles based on a favourable outcome. Hydrogels and dendrimers form a separate broader category in the sensitivity analysis; source-labelled hybrids remain outside that rule.

Material labels are coarse. The analysis does not contain a uniform polymer repeat-unit, molecular-weight distribution, block architecture or ligand-density representation. Therefore neither the core category nor these distribution summaries can define a new polymer synthesis target or explain a release/transport mechanism.

E Rank objective and uncertainty

Statistic	Estimate	95% lower	95% upper
Spearman tumour vs selectivity	0.5610	0.3958	0.7068
Median log10 selectivity	-1.0386	-1.2274	-0.8985
Top-quartile overlap fraction	0.4722	Not estimated	Not estimated

Table S6: Primary objective audit. Intervals use publication-cluster resampling.

id	study	strategy	tumor
338	178	Active	18.1400
361	194	Passive	9.1600
154	79	Passive	8.3700
337	178	Passive	7.7800
362	194	Passive	7.4600
155	79	Passive	6.5800
148	74	Active	5.3400
511	281	Passive	5.2200
213	109	Passive	4.8800
156	79	Passive	4.5900

Table S7: Highest tumour-uptake rows, disclosed as an objective illustration, not recommended materials. (Part 1 of 2.)

id	liver	log ratio
338	3.4500	0.7208
361	40.1500	-0.6418
154	11.9800	-0.1557
337	3.6500	0.3287
362	37.3300	-0.6993
155	14.2000	-0.3341
148	36.4100	-0.8337
511	5.3900	-0.0139
213	4.8500	0.0027
156	18.2300	-0.5990

Table S7: Highest tumour-uptake rows, disclosed as an objective illustration, not recommended materials. (Part 2 of 2.)

Cluster resampling draws source-publication blocks with replacement and carries all their included formulation records. A study with many formulations contributes many rows to formulation-level statistics, while the resampling unit remains a publication. The median-log-ratio interval and median-ratio point estimate are intentionally labelled separately; monotone transformation can be used if a ratio-scale interval is needed.

The top-quartile overlap uses source formulation IDs and a fixed ceil(n/4) list size. It is a transparent descriptive comparison, not an out-of-sample model-selection performance estimate. Tie sensitivity and alternative shortlist thresholds have not been exhaustively characterized.

F Targeting estimands and publication contrasts

Association estimand	log10 contrast	95% lower
Pooled formulation contrast	0.2346	-0.1116
Equal-weight publication contrast	0.2088	0.0914

Table S8: Repeated here to preserve the statistical estimands beside the study-level evidence. (Part 1 of 2.)

Association estimand	95% upper	Groups
Pooled formulation contrast	0.6054	72
Equal-weight publication contrast	0.3254	15

Table S8: Repeated here to preserve the statistical estimands beside the study-level evidence. (Part 2 of 2.)

Contrast index (sorted source group)	Active-minus-passive mean log10 ratio
1	0.4698
2	0.5711
3	0.1507
4	0.2325
5	0.1597
6	-0.2779
7	0.2683
8	0.3085
9	0.0000
10	0.3921
11	0.2769

Table S9: All overlapping-publication contrasts; no favourable-study selection. (Part 1 of 2.)

Contrast index (sorted source group)	Active-minus-passive mean log10 ratio
12	0.5359
13	-0.0102
14	-0.1353
15	0.1899

Table S9: All overlapping-publication contrasts; no favourable-study selection. (Part 2 of 2.)

The pooled contrast estimates a formulation-weighted association across the selected literature cohort. The within-publication contrast instead conditions on publications reporting both source targeting labels and gives each such publication equal weight. Their difference is not itself a formal test of confounding.

Even the within-publication contrast need not compare otherwise identical particles. Ligand addition may coincide with altered polymer chemistry, hydrodynamic diameter, charge, cargo, dose or tumour model. This audit does not harmonize animal randomization, blinding or study-level variances.

G Connected-reference grouping and split manifest

seed / role	Formul. n	groups
11 / reserved unused	22	15
11 / test	43	15
11 / train	77	42
29 / reserved unused	38	15
29 / test	24	15
29 / train	80	42
47 / reserved unused	30	15
47 / test	23	15
47 / train	89	42
71 / reserved unused	25	15
71 / test	31	15

Table S10: All executed primary partition roles. (Part 1 of 2.)

seed / role	Formul. n	groups
71 / train	86	42
101 / reserved unused	34	15
101 / test	23	15
101 / train	85	42

Table S10: All executed primary partition roles. (Part 2 of 2.)

The source reference 67/200 is not converted to missing and its formulations are not discarded. Instead reference IDs are graph nodes, a slash-linked record joins its nodes, and connected components define group labels. Thus records labelled 67, 200 and 67/200 receive the same group even when they occur far apart in the workbook.

The full mapping is visible in the analysis cohort and split manifest. Unit tests cover direct and transitive links, unknown-reference failure and absence of group overlap between roles. This addresses an identifier-integrity issue without asserting that all publications from one laboratory have been identified.

The reserved group is unused by this point-prediction study. It is not quietly added to training after looking at test scores. Changing that design in future work should produce a new recorded protocol version and new validation artifacts.

H Full predictive benchmark

Split / Model	MAE	Study-macro MAE
Random / Extra trees	0.5734	0.5832
Publication / Median	0.4637	0.4686
Publication / Ridge	0.4847	0.5117
Publication / Extra trees	0.5873	0.5920
Publication / No targeting	0.5874	0.5871
Publication / Shuffled targets	0.5372	0.5337
Publication / Physicochemical	0.5321	0.5558

Table S11: Mean metrics across all executed prediction conditions. (Part 1 of 2.)

Split / Model	R2
Random / Extra trees	-0.0538
Publication / Median	-0.0739
Publication / Ridge	-0.1328
Publication / Extra trees	-0.5850
Publication / No targeting	-0.5883
Publication / Shuffled targets	-0.4443
Publication / Physicochemical	-0.3466

Table S11: Mean metrics across all executed prediction conditions. (Part 2 of 2.)

seed	Test n	mae	Study MAE
11	43	0.5795	0.6552
29	24	0.7204	0.7111
47	23	0.5415	0.5475
71	31	0.5284	0.4765
101	23	0.5669	0.5695

Table S12: Primary tree scores by seed. (Part 1 of 2.)

seed	r2
11	-0.7940
29	-1.2438
47	-0.3480
71	-0.3298
101	-0.2092

Table S12: Primary tree scores by seed. (Part 2 of 2.)

MAE is measured in log10 ratio units. For perspective, a one-log10-unit absolute discrepancy represents a factor of ten in the ratio, but 10^MAE is

not the arithmetic mean multiplicative error. R^2 can be negative because the test mean is a stronger squared-error reference than this predictor.

The five repeated partitions overlap. Their dispersion is sensitivity to this splitting procedure, not an independent replication confidence interval. There is no test-set hyperparameter search, high-dimensional molecular embedding, GPU training or claim that these baselines are optimal.

I Time/material sensitivities and negative controls

hours / cohort	n	Pub. n
6 / polymeric	163	84
6 / polymer extended	187	99
24 / polymeric	142	72
24 / polymer extended	166	87
48 / polymeric	61	32
48 / polymer extended	77	43

Table S13: All prespecified descriptive sensitivities. (Part 1 of 3.)

hours / cohort	median tumor liver ratio	Rank correlation
6 / polymeric	0.0827	0.5141
6 / polymer extended	0.0797	0.5478
24 / polymeric	0.0915	0.5610
24 / polymer extended	0.0880	0.6043
48 / polymeric	0.1070	0.4757
48 / polymer extended	0.1032	0.6064

Table S13: All prespecified descriptive sensitivities. (Part 2 of 3.)

hours / cohort	Top-list overlap
6 / polymeric	0.4634
6 / polymer extended	0.5106
24 / polymeric	0.4722
24 / polymer extended	0.5238
48 / polymeric	0.4375
48 / polymer extended	0.5500

Table S13: All prespecified descriptive sensitivities. (Part 3 of 3.)

At each time, the paired curve must bracket that time and the endpoint must satisfy the same positivity and %ID range rules. Therefore the 6-, 24- and 48-hour cohorts differ. This is not a longitudinal analysis of an invariant formulation cohort and cannot separate kinetic changes from observation-selection differences.

The no-targeting model removes only the source Active/Passive feature. The physicochemical-only model uses hydrodynamic diameter, zeta potential and shape. A permuted-target control shuffles the training log-ratio target once per seed. These controls assess prediction dependence, not randomized biological targeting mechanisms.

The exact-time check is identical to the primary cohort and is retained as a transparent audit result. It does not multiply the effective sample size or justify presenting a separate replication success.

J Reproduction, environment and saved outputs

Instrumented property	Value
elapsed seconds	275.4892
sampled peak rss mb	226.2188
memory sampling seconds	0.1000
python	3.12.10
platform	Windows-11-10.0.26300-SP0
cpu threads	1

Table S14: Analysis-only measured compute record.

From the workspace root, install the pinned requirements into the local environment and run fetch_sources.py followed by 02_polymer_targeting\run.py. To rebuild both papers, run both project pipelines first and then build_manuscripts.py. The common root utilities are required; the project folders are scientifically separate, not standalone dependency-free software packages.

Saved result	Purpose
analysis dataset.csv	Primary polymer endpoint cohort and reference groups
all material endpoints.csv	Paired 24-h endpoints before material filtering
data audit.json	Counts, missingness, exclusions, source hash
effects.json	Ranks, bootstrap intervals and targeting contrasts

Table S15: Machine-readable evidence map. (Part 1 of 2.)

Saved result	Purpose
metrics.csv / predictions.csv	Every evaluated split/model score and tree prediction
splits.csv	Identifiers and roles for every seed
sensitivity.csv / protocol.json	Declared sensitivity cohorts and frozen settings

Table S15: Machine-readable evidence map. (Part 2 of 2.)

No clinical or experimental result is generated by building the PDF. The manuscript reads measured result files; a change in source data requires rerunning analysis. Test outputs and document validation are separate from scientific peer review and should not be confused with confirmation of the biological interpretation.

K External validation design

Design element	Proposed evaluation
Comparison	Composition-matched active and passive carriers
Exposure	Separate measurements of carrier and free drug
Distribution	Paired multiorgan observations at common times
Benefit	Tumour response evaluated independently of accumulation

Table S16: Design for independent evaluation of a selectivity objective. (Part 1 of 2.)

Design element	Proposed evaluation
Risk	Organ-specific toxicity assessed separately from selectivity

Table S16: Design for independent evaluation of a selectivity objective. (Part 2 of 2.)

An independent study could compare composition-matched active/passive polymer carriers in a genuinely independent data resource, quantify free-drug and carrier distributions separately, and test whether a multiorgan objective predicts a meaningful therapeutic outcome. Such validation has not been performed here; the present results describe published biodistribution observations.

L Extension protocol and information boundaries

The extension is exploratory and follows the completed primary analysis. It does not retroactively label the additional model configurations or robustness questions as preregistered. The primary cohort, source exclusions, time units and publication partitions are frozen. Additional configurations use the same prediction target and the same original holdouts. No test-set search selects a new primary model or replaces the primary scores.

Setting	Retained predictors	Learner
Trees 128 leaf3	6	Extra trees
Ridge alpha1	6	Ridge
Ridge alpha10	6	Ridge
Ridge alpha100	6	Ridge
Trees 128 leaf1	6	Extra trees
Trees 128 leaf8	6	Extra trees
Trees 32 leaf3	6	Extra trees
Without targeting	5	Extra trees
Without tumour label	5	Extra trees
Without size	5	Extra trees
Without zeta	5	Extra trees

Table S17: Complete executed extension configuration grid. (Part 1 of 4.)

Setting	Retained predictors	Learner
Without shape	5	Extra trees

Table S17: Complete executed extension configuration grid. (Part 2 of 4.)

Setting	Penalty / leaf	Trees
Trees 128 leaf3	3	128
Ridge alpha1	1	0
Ridge alpha10	10	0
Ridge alpha100	100	0
Trees 128 leaf1	1	128
Trees 128 leaf8	8	128
Trees 32 leaf3	3	32
Without targeting	3	128
Without tumour label	3	128
Without size	3	128
Without zeta	3	128

Table S17: Complete executed extension configuration grid. (Part 3 of 4.)

Setting	Penalty / leaf	Trees
Without shape	3	128

Table S17: Complete executed extension configuration grid. (Part 4 of 4.)

Preprocessing remains training-only. Numeric variables are median imputed with missingness indicators and standardized. Categorical variables are imputed from the training mode and one-hot encoded with unknown categories ignored at prediction time. The constant source core category in the polymer cohort is retained for compatibility with the original pipeline rather than presented as a source of discriminative information.

The extension reads saved numeric inputs using round-trip floating-point parsing. This matters for exact reproducibility: rounding a continuous descriptor before standardization can alter a randomized tree split. Before interpreting the grid, the baseline is refitted and its five publication-split MAEs are checked against the original saved results to tight numerical tolerance. The extension records hashes of the frozen cohort, primary metrics and partition manifest.

All estimators use one worker. Instrumentation samples process memory every 0.1 seconds and excludes dependency installation and document compilation. Safety checks stop model-grid execution before 110 minutes or 3.8 GB sampled process memory. The combined measured primary and extension runtime is checked against the two-hour analysis budget.

M Complete model-grid and paired ablation results

Every row below identifies an outer seed and an executed configuration. Formulation-weighted MAE gives equal weight to eligible formulations. Group MAE first averages errors within a publication group and then gives groups equal weight. R^2 uses the same test-target comparator as the primary benchmark. A configuration can improve one weighting scheme while worsening another when group sizes and error patterns differ.

Seed / setting	mae	Study MAE	r2
11 / Trees 128 leaf3	0.5795	0.6552	-0.7940
11 / Ridge alpha1	0.4982	0.5481	-0.4059
11 / Ridge alpha10	0.4092	0.4923	-0.0008
11 / Ridge alpha100	0.4194	0.4855	-0.0609
11 / Trees 128 leaf1	0.5898	0.5910	-0.9868
11 / Trees 128 leaf8	0.4826	0.5691	-0.4113
11 / Trees 32 leaf3	0.6128	0.6809	-0.9234
11 / Without targeting	0.6008	0.6578	-0.8787
11 / Without tumour label	0.5102	0.5914	-0.5303
11 / Without size	0.5239	0.5732	-0.4726
11 / Without zeta	0.5013	0.5756	-0.6182

Table S18: All per-partition model and refit-ablation scores. (Part 1 of 6.)

Seed / setting	mae	Study MAE	r2
11 / Without shape	0.5265	0.5902	-0.5610
29 / Trees 128 leaf3	0.7204	0.7111	-1.2438
29 / Ridge alpha1	0.6635	0.7039	-0.9306
29 / Ridge alpha10	0.6110	0.6564	-0.7641
29 / Ridge alpha100	0.5029	0.5671	-0.2368
29 / Trees 128 leaf1	0.6661	0.7129	-1.0160
29 / Trees 128 leaf8	0.4335	0.4975	0.0800
29 / Trees 32 leaf3	0.7247	0.7134	-1.2784
29 / Without targeting	0.7295	0.7250	-1.2717
29 / Without tumour label	0.6939	0.7077	-1.1528
29 / Without size	0.5423	0.5596	-0.3046

Table S18: All per-partition model and refit-ablation scores. (Part 2 of 6.)

Seed / setting	mae	Study MAE	r2
29 / Without zeta	0.6364	0.6565	-0.9041
29 / Without shape	0.7331	0.7370	-1.2445
47 / Trees 128 leaf3	0.5415	0.5475	-0.3480
47 / Ridge alpha1	0.4913	0.5398	-0.2594
47 / Ridge alpha10	0.4507	0.4955	-0.1173
47 / Ridge alpha100	0.4449	0.4798	-0.0561
47 / Trees 128 leaf1	0.4713	0.4671	-0.2849
47 / Trees 128 leaf8	0.4470	0.4933	-0.0739
47 / Trees 32 leaf3	0.5389	0.5450	-0.3248
47 / Without targeting	0.5428	0.5420	-0.3704
47 / Without tumour label	0.4997	0.5326	-0.2687

Table S18: All per-partition model and refit-ablation scores. (Part 3 of 6.)

Seed / setting	mae	Study MAE	r2
47 / Without size	0.5940	0.5722	-0.4760
47 / Without zeta	0.4845	0.4803	-0.0993
47 / Without shape	0.4995	0.5286	-0.2964
71 / Trees 128 leaf3	0.5284	0.4765	-0.3298
71 / Ridge alpha1	0.4443	0.4221	0.2656
71 / Ridge alpha10	0.4723	0.4242	0.0421
71 / Ridge alpha100	0.4663	0.4104	-0.0516
71 / Trees 128 leaf1	0.3948	0.3916	0.4489
71 / Trees 128 leaf8	0.5064	0.4407	0.0179
71 / Trees 32 leaf3	0.5191	0.4722	-0.3290
71 / Without targeting	0.5095	0.4511	-0.2885

Table S18: All per-partition model and refit-ablation scores. (Part 4 of 6.)

Seed / setting	mae	Study MAE	r2
71 / Without tumour label	0.4649	0.4558	0.1456
71 / Without size	0.5318	0.4687	-0.4104
71 / Without zeta	0.4937	0.4456	-0.2220
71 / Without shape	0.4989	0.4464	-0.2738
101 / Trees 128 leaf3	0.5669	0.5695	-0.2092
101 / Ridge alpha1	0.5336	0.5718	-0.0091
101 / Ridge alpha10	0.4804	0.4901	0.1763
101 / Ridge alpha100	0.4969	0.4689	0.0874
101 / Trees 128 leaf1	0.8335	0.8282	-1.6000
101 / Trees 128 leaf8	0.5940	0.5845	-0.2210
101 / Trees 32 leaf3	0.5667	0.5700	-0.2133

Table S18: All per-partition model and refit-ablation scores. (Part 5 of 6.)

Seed / setting	mae	Study MAE	r2
101 / Without targeting	0.5543	0.5597	-0.1321
101 / Without tumour label	0.5296	0.5728	0.0020
101 / Without size	0.5537	0.5479	-0.1715
101 / Without zeta	0.5745	0.5644	-0.2881
101 / Without shape	0.5916	0.5755	-0.3070

Table S18: All per-partition model and refit-ablation scores. (Part 6 of 6.)

configuration	mean	min	max
Ridge alpha1	-0.0612	-0.0841	-0.0332
Ridge alpha10	-0.1026	-0.1703	-0.0560
Ridge alpha100	-0.1212	-0.2174	-0.0621
Trees 128 leaf1	0.0038	-0.1335	0.2667
Trees 128 leaf3	0.0000	0.0000	0.0000
Trees 128 leaf8	-0.0946	-0.2869	0.0271
Trees 32 leaf3	0.0051	-0.0092	0.0334
Without shape	-0.0174	-0.0530	0.0247
Without size	-0.0382	-0.1780	0.0524
Without targeting	0.0001	-0.0188	0.0213
Without tumour label	-0.0477	-0.0693	-0.0264

Table S19: Paired MAE difference from the original tree baseline; negative means lower error. (Part 1 of 2.)

configuration	mean	min	max
Without zeta	-0.0492	-0.0839	0.0076

Table S19: Paired MAE difference from the original tree baseline; negative means lower error. (Part 2 of 2.)

The paired differences use the same held-out observations within each seed. Their range shows whether a comparison changes direction across the reported partitions. It is not a confidence interval for population risk. The repeated partitions share publications, and the configuration grid was evaluated after primary-result inspection. Consequently neither the smallest mean error nor a uniformly negative five-seed difference supplies a confirmatory discovery claim.

configuration	target	mae	r2
Ridge alpha1	log ratio	0.5262	-0.2679
Ridge alpha10	log ratio	0.4847	-0.1328
Ridge alpha100	log ratio	0.4661	-0.0636
Trees 128 leaf1	log ratio	0.5911	-0.6877
Trees 128 leaf3	log ratio	0.5873	-0.5850
Trees 128 leaf8	log ratio	0.4927	-0.1216
Trees 32 leaf3	log ratio	0.5925	-0.6138
Without shape	log ratio	0.5699	-0.5365
Without size	log ratio	0.5491	-0.3670
Without targeting	log ratio	0.5874	-0.5883
Without tumour label	log ratio	0.5397	-0.3608

Table S20: Target-specific mean MAE and R^2 for every extension configuration. (Part 1 of 2.)

configuration	target	mae	r2
Without zeta	log ratio	0.5381	-0.4264

Table S20: Target-specific mean MAE and R^2 for every extension configuration. (Part 2 of 2.)

For polymer selectivity there is a single log-ratio target, so the endpoint table coincides with the configuration summary. No endpoint is selected retrospectively to make a model look better.

N Nested publication learning curves

A deterministic seed-specific permutation orders the original training publication groups. The four learning subsets are nested prefixes of this ordering, with the number of groups rounded upward. All formulations belonging to a retained group are kept. Calibration or reserved groups never enter training, and the original test set remains unchanged at every training fraction.

seed	fraction	train n	train groups
11	0.2500	22	11
11	0.5000	46	21
11	0.7500	61	32
11	1.0000	77	42
29	0.2500	18	11
29	0.5000	37	21
29	0.7500	63	32
29	1.0000	80	42
47	0.2500	17	11
47	0.5000	35	21
47	0.7500	59	32

Table S21: Every training-fraction and outer-partition result. (Part 1 of 4.)

seed	fraction	train n	train groups
47	1.0000	89	42
71	0.2500	29	11
71	0.5000	51	21
71	0.7500	74	32
71	1.0000	86	42
101	0.2500	25	11
101	0.5000	47	21
101	0.7500	71	32
101	1.0000	85	42

Table S21: Every training-fraction and outer-partition result. (Part 2 of 4.)

seed	mae	Study MAE	r2
11	0.4881	0.5881	-0.3652
11	0.4605	0.5728	-0.2958
11	0.5475	0.6202	-0.7061
11	0.5795	0.6552	-0.7940
29	0.5124	0.5758	-0.1998
29	0.4888	0.5403	-0.0026
29	0.4666	0.5314	-0.0332
29	0.7204	0.7111	-1.2438
47	0.5254	0.5442	-0.2704
47	0.5751	0.5295	-0.4410
47	0.5182	0.5190	-0.2551

Table S21: Every training-fraction and outer-partition result. (Part 3 of 4.)

seed	mae	Study MAE	r2
47	0.5415	0.5475	-0.3480
71	0.4878	0.4226	-0.2988
71	0.4997	0.4226	-0.1081
71	0.5132	0.4354	-0.2596
71	0.5284	0.4765	-0.3298
101	0.5264	0.4737	-0.1020
101	0.4848	0.4380	0.0731
101	0.5984	0.5352	-0.4906
101	0.5669	0.5695	-0.2092

Table S21: Every training-fraction and outer-partition result. (Part 4 of 4.)

The 100% subset is exactly the original training set, and its tree prediction score must reproduce the extension baseline. The saved learning manifest contains every included formulation identifier, group identifier, fraction and seed. Automated checks require nested identifier sets, complete groups, no duplicates and no overlap with held-out publications.

Learning curves at this scale need not decrease monotonically. Each added block changes the empirical training mixture, and the fixed estimator is not reoptimized at each size. A flatter curve can reflect missing information or limited diversity as well as estimator bias. Four exploratory fractions are insufficient to support a reliable asymptotic learning exponent or a power calculation for a future drug-development study.

O Descriptor support across unseen publications

Numeric support is the closed interval between the minimum and maximum nonmissing training values. Categorical support is the set of observed nonmissing training labels. A test value outside the

corresponding set is flagged. Missing test values are recorded independently and remain subject to the training-only imputation rule.

seed	feature	support type
11	size nm	Outside training range
11	zeta mv	Outside training range
11	strategy	Unseen category
11	shape	Unseen category
11	core	Unseen category
11	tumor cell	Unseen category
29	size nm	Outside training range
29	zeta mv	Outside training range

Table S22: Complete feature-by-partition support and missingness audit. (Part 1 of 10.)

seed	feature	support type
29	strategy	Unseen category
29	shape	Unseen category
29	core	Unseen category
29	tumor cell	Unseen category
47	size nm	Outside training range
47	zeta mv	Outside training range
47	strategy	Unseen category
47	shape	Unseen category
47	core	Unseen category
47	tumor cell	Unseen category

Table S22: Complete feature-by-partition support and missingness audit. (Part 2 of 10.)

seed	feature	support type
71	size nm	Outside training range
71	zeta mv	Outside training range
71	strategy	Unseen category
71	shape	Unseen category
71	core	Unseen category
71	tumor cell	Unseen category
101	size nm	Outside training range
101	zeta mv	Outside training range

Table S22: Complete feature-by-partition support and missingness audit. (Part 3 of 10.)

seed	feature	support type
101	strategy	Unseen category
101	shape	Unseen category
101	core	Unseen category
101	tumor cell	Unseen category

Table S22: Complete feature-by-partition support and missingness audit. (Part 4 of 10.)

seed	test n	unsupported n
11	43	1
11	43	0
11	43	0
11	43	0
11	43	0
11	43	25
29	24	2
29	24	1
29	24	0
29	24	0
29	24	0

Table S22: Complete feature-by-partition support and missingness audit. (Part 5 of 10.)

seed	test n	unsupported n
29	24	12
47	23	0
47	23	0
47	23	0
47	23	0
47	23	0
47	23	17
71	31	0
71	31	2
71	31	0
71	31	0

Table S22: Complete feature-by-partition support and missingness audit. (Part 6 of 10.)

seed	test n	unsupported n
71	31	0
71	31	13
101	23	0
101	23	0
101	23	0
101	23	1
101	23	0
101	23	10

Table S22: Complete feature-by-partition support and missingness audit. (Part 7 of 10.)

seed	unsupported fraction	missing fraction
11	0.0233	0.0698
11	0.0000	0.0698
11	0.0000	0.0000
11	0.0000	0.0000
11	0.0000	0.0000
11	0.5814	0.0000
29	0.0833	0.1667
29	0.0417	0.0000
29	0.0000	0.0000
29	0.0000	0.0000
29	0.0000	0.0000

Table S22: Complete feature-by-partition support and missingness audit. (Part 8 of 10.)

seed	unsupported fraction	missing fraction
29	0.5000	0.0000
47	0.0000	0.0000
47	0.0000	0.0870
47	0.0000	0.0000
47	0.0000	0.0000
47	0.0000	0.0000
47	0.7391	0.0000
71	0.0000	0.0000
71	0.0645	0.0968
71	0.0000	0.0000
71	0.0000	0.0000

Table S22: Complete feature-by-partition support and missingness audit. (Part 9 of 10.)

seed	unsupported fraction	missing fraction
71	0.0000	0.0000
71	0.4194	0.0000
101	0.0000	0.0435
101	0.0000	0.0870
101	0.0000	0.0000
101	0.0435	0.0000
101	0.0000	0.0000
101	0.4348	0.0000

Table S22: Complete feature-by-partition support and missingness audit. (Part 10 of 10.)

feature	unsupported fraction	missing fraction
core	0.0000	0.0000
shape	0.0087	0.0000
size nm	0.0213	0.0560
strategy	0.0000	0.0000
tumor cell	0.5349	0.0000
zeta mv	0.0212	0.0681

Table S23: Mean unsupported and missing fractions across the five partitions.

The audit does not use target values. Its simple univariate definition is deliberately inspectable, but it cannot detect every joint covariate shift. A test formulation may lie inside each marginal range while combining descriptors in a way never observed during training. Likewise, a slightly out-of-range value is not necessarily a more consequential shift than an unrecorded change in assay medium or polymer architecture.

Unseen tumour-cell or shape labels are not automatically evidence of an invalid measurement. They show that the fitted categorical representation cannot recover a category-specific effect from training. Ignoring unseen one-hot categories avoids fitting on test data but does not solve transportability. The distinction between a working encoder and a biologically informative representation remains essential.

P Held-out permutation diagnostics

After fitting the original tree baseline within each outer partition, we independently permute one test-set descriptor at a time. Five deterministic shuffles are applied per feature and seed. The diagnostic is the shuffled MAE minus the unperturbed MAE. It is computed on held-out observations but is not used to choose features for a subsequently reported test score.

feature	mean	std
core	0.0000	0.0000
shape	0.0015	0.0072
size nm	-0.0378	0.0411
strategy	-0.0066	0.0114
tumor cell	-0.0049	0.0350
zeta mv	-0.0322	0.0316

Table S24: Mean and SD of held-out feature-permutation MAE increases.

seed	feature	mae increase
11	core	0.0000
11	shape	0.0000
11	size nm	-0.0280
11	strategy	-0.0012
11	tumor cell	0.0255
11	zeta mv	-0.0263
29	core	0.0000
29	shape	0.0000
29	size nm	-0.1073
29	strategy	-0.0012
29	tumor cell	0.0094

Table S25: Every partition-specific permutation diagnostic, averaged over five shuffles. (Part 1 of 3.)

seed	feature	mae increase
29	zeta mv	-0.0488
47	core	0.0000
47	shape	0.0000
47	size nm	-0.0354
47	strategy	-0.0098
47	tumor cell	0.0018
47	zeta mv	-0.0587
71	core	0.0000
71	shape	0.0000
71	size nm	-0.0120
71	strategy	-0.0037

Table S25: Every partition-specific permutation diagnostic, averaged over five shuffles. (Part 2 of 3.)

seed	feature	mae increase
71	tumor cell	-0.0261
71	zeta mv	-0.0304
101	core	0.0000
101	shape	0.0077
101	size nm	-0.0061
101	strategy	-0.0172
101	tumor cell	-0.0349
101	zeta mv	0.0033

Table S25: Every partition-specific permutation diagnostic, averaged over five shuffles. (Part 3 of 3.)

A positive value means the fitted predictor performed worse after shuffling, not that the feature caused improved delivery. Correlated predictors can substitute for each other, and shuffling can create implausible combinations of drug, polymer and assay descriptors. A negative value is possible under weak prediction or finite-sample noise and is retained rather than truncated.

The reported SD pools exploratory perturbation results and is not an independent-publication confidence interval. Refit ablations ask a different question by permitting the model to adapt to removed information. Agreement between the two diagnostics can support interpretation of the fitted representation, but neither identifies a mechanistic contribution or justifies changing a material without experimental validation.

Q Objective-weight and shortlist-size grid

The objective grid crosses five shortlist fractions with six liver penalties. Every cell uses the same 142 paired observations. A shortlist is constructed by descending score with the source-row order resolving exact ties through the same deterministic table operation. The uptake and ratio reference shortlists use that same requested list length. No cell is labelled the optimal screening rule.

fraction	liver weight	top n	uptake overlap
0.1000	0.0000	15	1.0000
0.1000	0.2500	15	0.8667
0.1000	0.5000	15	0.5333
0.1000	1.0000	15	0.3333
0.1000	1.5000	15	0.1333
0.1000	2.0000	15	0.1333
0.2000	0.0000	29	1.0000
0.2000	0.2500	29	0.7931
0.2000	0.5000	29	0.7241
0.2000	1.0000	29	0.3793
0.2000	1.5000	29	0.2414

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 1 of 9.)

fraction	liver weight	top n	uptake overlap
0.2000	2.0000	29	0.2069
0.2500	0.0000	36	1.0000
0.2500	0.2500	36	0.8889
0.2500	0.5000	36	0.6944
0.2500	1.0000	36	0.4722
0.2500	1.5000	36	0.3056
0.2500	2.0000	36	0.2222
0.3300	0.0000	47	1.0000
0.3300	0.2500	47	0.8298
0.3300	0.5000	47	0.7234
0.3300	1.0000	47	0.5106

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 2 of 9.)

fraction	liver weight	top n	uptake overlap
0.3300	1.5000	47	0.3617
0.3300	2.0000	47	0.2766
0.5000	0.0000	71	1.0000
0.5000	0.2500	71	0.9296
0.5000	0.5000	71	0.8592
0.5000	1.0000	71	0.7606
0.5000	1.5000	71	0.6479
0.5000	2.0000	71	0.5775

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 3 of 9.)

fraction	ratio overlap	median tumor	median liver
0.1000	0.3333	5.2200	14.2000
0.1000	0.4000	5.2200	14.0200
0.1000	0.7333	4.8800	3.6500
0.1000	1.0000	1.0100	0.4400
0.1000	0.8000	0.5600	0.3300
0.1000	0.7333	0.5200	0.2600
0.2000	0.3793	3.5300	14.3500
0.2000	0.4828	3.5300	11.9800
0.2000	0.5862	3.0100	7.7400
0.2000	1.0000	1.0100	1.3400
0.2000	0.8621	0.7000	0.7600

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 4 of 9.)

fraction	ratio overlap	median tumor	median liver
0.2000	0.8276	0.6200	0.7600
0.2500	0.4722	2.8500	15.2500
0.2500	0.5833	2.8500	13.7300
0.2500	0.7778	2.8100	7.0150
0.2500	1.0000	1.4600	3.2800
0.2500	0.8333	0.7700	1.3000
0.2500	0.7222	0.5800	1.0300
0.3300	0.5106	2.5900	17.4500
0.3300	0.6809	2.5900	13.5200
0.3300	0.7872	2.5900	8.3500
0.3300	1.0000	1.6200	3.8800

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 5 of 9.)

fraction	ratio overlap	median tumor	median liver
0.3300	0.8511	0.8600	1.8800
0.3300	0.7660	0.7000	1.3400
0.5000	0.7606	1.8800	12.9300
0.5000	0.8310	1.8800	11.9800
0.5000	0.9014	1.8800	9.4000
0.5000	1.0000	1.6200	6.8400
0.5000	0.8732	1.0300	4.2400
0.5000	0.8028	0.8600	3.8800

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 6 of 9.)

fraction	rank vs uptake
0.1000	1.0000
0.1000	0.9667
0.1000	0.8713
0.1000	0.5610
0.1000	0.2927
0.1000	0.1394
0.2000	1.0000
0.2000	0.9667
0.2000	0.8713
0.2000	0.5610
0.2000	0.2927

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 7 of 9.)

fraction	rank vs uptake
0.2000	0.1394
0.2500	1.0000
0.2500	0.9667
0.2500	0.8713
0.2500	0.5610
0.2500	0.2927
0.2500	0.1394
0.3300	1.0000
0.3300	0.9667
0.3300	0.8713
0.3300	0.5610

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 8 of 9.)

fraction	rank vs uptake
0.3300	0.2927
0.3300	0.1394
0.5000	1.0000
0.5000	0.9667
0.5000	0.8713
0.5000	0.5610
0.5000	0.2927
0.5000	0.1394

Table S26: Complete objective-weight and shortlist-fraction grid. (Part 9 of 9.)

For lambda greater than one, the score penalizes liver accumulation more strongly than the log ratio does. For lambda between zero and one it moves toward tumour accumulation alone. These changes encode preferences, not additional information or a new treatment mechanism. Median organ accumulation among selected rows is included so that an apparently favourable ratio is not read without its numerator and denominator.

id	tumor	liver	log ratio
229	0.0100	0.0100	0.0000
338	18.1400	3.4500	0.7208
407	0.8600	0.1000	0.9345
408	0.5200	0.0900	0.7618
409	1.0400	0.4200	0.3938
435	2.8600	1.3400	0.3293

Table S27: Recorded paired-organ Pareto-frontier rows; not recommended therapies. (Part 1 of 2.)

id	strategy
229	Passive
338	Active
407	Active
408	Active
409	Active
435	Passive

Table S27: Recorded paired-organ Pareto-frontier rows; not recommended therapies. (Part 2 of 2.)

The frontier maximizes tumour accumulation and minimizes liver accumulation simultaneously. Tied observations remain on the frontier when neither strictly dominates the other. Because the comparison pools heterogeneous experiments, frontier membership is descriptive and may change after harmonizing dose, tumour model, sampling conditions or measurement uncertainty.

R Publication-omission influence analysis

One connected publication-reference group is removed per iteration. All remaining observations keep their original organ measurements and targeting categories. We then recompute the pooled active-minus-passive mean, the equal-weight mean of overlapping-group contrasts, the median log ratio and the uptake/selectivity rank correlation. This deterministic influence analysis differs from repeatedly resampling groups with replacement.

omitted group	remaining n	rank correlation
105	140	0.5878
109	141	0.5540
113	138	0.5732
114	141	0.5526
116	141	0.5894
118	140	0.5591
119	141	0.5577
120	141	0.5591
121	140	0.5602
127	141	0.5616
128	141	0.5520

Table S28: Complete leave-one-reference-group-out influence audit. (Part 1 of 21.)

omitted group	remaining n	rank correlation
130	137	0.5708
131	141	0.5645
133	141	0.5791
138	140	0.5500
142	141	0.5731
143	141	0.5616
147	138	0.5514
148	141	0.5681
149	141	0.5593
151	141	0.5563
152	140	0.5611

Table S28: Complete leave-one-reference-group-out influence audit. (Part 2 of 21.)

omitted group	remaining n	rank correlation
160	141	0.5566
161	140	0.5525
163	141	0.5622
166	141	0.5594
167	140	0.5625
169	140	0.5829
173	141	0.5599
174	139	0.5661
175	140	0.5602
178	140	0.5441
180	141	0.5615

Table S28: Complete leave-one-reference-group-out influence audit. (Part 3 of 21.)

omitted group	remaining n	rank correlation
184	141	0.5621
186	141	0.5801
189	141	0.5663
191	141	0.5595
194	140	0.5570
197	130	0.5625
205	141	0.5631
207	139	0.5543
214	138	0.5932
225	141	0.5619
228	140	0.5572

Table S28: Complete leave-one-reference-group-out influence audit. (Part 4 of 21.)

omitted group	remaining n	rank correlation
232	140	0.5572
236	140	0.5585
241	140	0.5445
242	141	0.5540
245	141	0.5576
247	141	0.5594
256	141	0.5647
257	141	0.5605
261	140	0.5574
273	140	0.5570
281	141	0.5545

Table S28: Complete leave-one-reference-group-out influence audit. (Part 5 of 21.)

omitted group	remaining n	rank correlation
297	141	0.5539
62	141	0.5613
63	140	0.5599
65	140	0.5697
67+200	138	0.5610
69	138	0.5589
70	139	0.5576
71	140	0.5564
72	136	0.5545
74	141	0.5615
75	141	0.5616

Table S28: Complete leave-one-reference-group-out influence audit. (Part 6 of 21.)

omitted group	remaining n	rank correlation
76	141	0.5596
77	140	0.5526
78	141	0.5615
79	134	0.5264
80	141	0.5615
81	140	0.5553

Table S28: Complete leave-one-reference-group-out influence audit. (Part 7 of 21.)

omitted group	median log ratio	within contrast
105	-1.0386	0.2088
109	-1.0476	0.2088
113	-1.0642	0.1902
114	-1.0296	0.2088
116	-1.0476	0.2088
118	-1.0386	0.1829
119	-1.0476	0.2088
120	-1.0296	0.2088
121	-1.0555	0.2130
127	-1.0476	0.2088
128	-1.0296	0.2088

Table S28: Complete leave-one-reference-group-out influence audit. (Part 8 of 21.)

omitted group	median log ratio	within contrast
130	-1.0100	0.2088
131	-1.0476	0.2088
133	-1.0476	0.2088
138	-1.0239	0.2071
142	-1.0476	0.2088
143	-1.0476	0.2088
147	-1.0141	0.2123
148	-1.0476	0.2088
149	-1.0296	0.2088
151	-1.0476	0.2088
152	-1.0386	0.2436

Table S28: Complete leave-one-reference-group-out influence audit. (Part 9 of 21.)

omitted group	median log ratio	within contrast
160	-1.0296	0.2088
161	-1.0239	0.2046
163	-1.0296	0.2088
166	-1.0296	0.2088
167	-1.0465	0.2017
169	-1.0555	0.2237
173	-1.0296	0.2088
174	-1.0476	0.2088
175	-1.0555	0.2088
178	-1.0555	0.1957
180	-1.0476	0.2088

Table S28: Complete leave-one-reference-group-out influence audit. (Part 10 of 21.)

omitted group	median log ratio	within contrast
184	-1.0476	0.2088
186	-1.0476	0.2088
189	-1.0296	0.2088
191	-1.0296	0.2088
194	-1.0555	0.2088
197	-1.1113	0.2088
205	-1.0296	0.2088
207	-1.0634	0.2088
214	-1.0642	0.2088
225	-1.0476	0.2088
228	-1.0386	0.2088

Table S28: Complete leave-one-reference-group-out influence audit. (Part 11 of 21.)

omitted group	median log ratio	within contrast
232	-1.0555	0.2088
236	-1.0239	0.2088
241	-1.0239	0.2088
242	-1.0296	0.2088
245	-1.0296	0.2088
247	-1.0476	0.2088
256	-1.0296	0.2088
257	-1.0296	0.2088
261	-1.0239	0.2039
273	-1.0239	0.2088
281	-1.0476	0.2088

Table S28: Complete leave-one-reference-group-out influence audit. (Part 12 of 21.)

omitted group	median log ratio	within contrast
297	-1.0296	0.2088
62	-1.0476	0.2088
63	-1.0386	0.1854
65	-1.0386	0.2088
67+200	-1.0386	0.2088
69	-1.0239	0.2088
70	-1.0182	0.2244
71	-1.0239	0.2088
72	-1.0056	0.2334
74	-1.0476	0.2088
75	-1.0476	0.2088

Table S28: Complete leave-one-reference-group-out influence audit. (Part 13 of 21.)

omitted group	median log ratio	within contrast
76	-1.0476	0.2088
77	-1.0239	0.2102
78	-1.0296	0.2088
79	-1.0239	0.2088
80	-1.0476	0.2088
81	-1.0555	0.2088

Table S28: Complete leave-one-reference-group-out influence audit. (Part 14 of 21.)

omitted group	matched groups	pooled contrast
105	15	0.2403
109	15	0.2454
113	14	0.1981
114	15	0.2233
116	15	0.2454
118	14	0.2336
119	15	0.2410
120	15	0.2325
121	14	0.2306
127	15	0.2389
128	15	0.2145

Table S28: Complete leave-one-reference-group-out influence audit. (Part 15 of 21.)

omitted group	matched groups	pooled contrast
130	15	0.2131
131	15	0.2359
133	15	0.2404
138	14	0.2508
142	15	0.2201
143	15	0.2382
147	14	0.2519
148	15	0.2500
149	15	0.2326
151	15	0.2430
152	14	0.2445

Table S28: Complete leave-one-reference-group-out influence audit. (Part 16 of 21.)

omitted group	matched groups	pooled contrast
160	15	0.2553
161	14	0.2477
163	15	0.2344
166	15	0.2322
167	14	0.2318
169	14	0.2355
173	15	0.2348
174	15	0.2579
175	15	0.2394
178	14	0.2069
180	15	0.2373

Table S28: Complete leave-one-reference-group-out influence audit. (Part 17 of 21.)

omitted group	matched groups	pooled contrast
184	15	0.2411
186	15	0.2217
189	15	0.2340
191	15	0.2331
194	15	0.2431
197	15	0.2588
205	15	0.2346
207	15	0.2183
214	15	0.0733
225	15	0.2410
228	15	0.2282

Table S28: Complete leave-one-reference-group-out influence audit. (Part 18 of 21.)

omitted group	matched groups	pooled contrast
232	15	0.2578
236	15	0.2266
241	15	0.2015
242	15	0.2294
245	15	0.2491
247	15	0.2391
256	15	0.2515
257	15	0.2272
261	14	0.2387
273	15	0.2306
281	15	0.2453

Table S28: Complete leave-one-reference-group-out influence audit. (Part 19 of 21.)

omitted group	matched groups	pooled contrast
297	15	0.2540
62	15	0.2367
63	14	0.2330
65	15	0.2353
67+200	15	0.2273
69	15	0.2280
70	14	0.2512
71	15	0.2277
72	14	0.3066
74	15	0.2338
75	15	0.2356

Table S28: Complete leave-one-reference-group-out influence audit. (Part 20 of 21.)

omitted group	matched groups	pooled contrast
76	15	0.2398
77	14	0.2529
78	15	0.2266
79	15	0.2176
80	15	0.2358
81	15	0.2447

Table S28: Complete leave-one-reference-group-out influence audit. (Part 21 of 21.)

A change in the pooled contrast can arise when a group supplies many observations from one targeting category. The within-group summary responds to a different intervention: omitting a group with both categories removes one contrast from an equal-weight average. Omission of a group containing only one category leaves that within-group average unchanged. This distinction explains why the two influence profiles need not track each other.

The minimum and maximum omission values are not a 95% interval and are not adjusted for publication selection or measurement error. They describe sensitivity to the finite source collection. A stable omission profile would not remove confounding within each publication or establish transportability to a new laboratory. The full audit is retained rather than showing only the most favourable omissions.

S Covariate balance and coupling diagnostics

For each numerical covariate, the pooled standardized difference divides the active-minus-passive mean difference by the square root of the average of the two sample variances. It is a descriptive balance measure, not a propensity score. The within-publication numerical contrast uses only groups with both categories and available numerical observations. Thus its eligible groups can differ between size and zeta potential.

feature	active n	passive n
size nm	35	96
zeta mv	35	95

Table S29: Numerical covariate balance by targeting category. (Part 1 of 4.)

feature	active mean	passive mean
size nm	130.8497	124.7889
zeta mv	-12.2680	-5.2359

Table S29: Numerical covariate balance by targeting category. (Part 2 of 4.)

feature	standardized difference	overlap groups
size nm	0.0774	12
zeta mv	-0.4096	12

Table S29: Numerical covariate balance by targeting category. (Part 3 of 4.)

feature	within mean difference
size nm	16.8923
zeta mv	-2.2022

Table S29: Numerical covariate balance by targeting category. (Part 4 of 4.)

cohort	n	groups
Primary	142	72
Complete size and zeta	119	57
Liver at least 0.01	142	72
Liver at least 0.1	139	70
Liver at least 1	124	61

Table S30: Complete physicochemical-case and minimum-liver sensitivity results. (Part 1 of 3.)

cohort	rank correlation	median log ratio
Primary	0.5610	-1.0386
Complete size and zeta	0.4689	-1.0100
Liver at least 0.01	0.5610	-1.0386
Liver at least 0.1	0.5915	-1.0476
Liver at least 1	0.7669	-1.1034

Table S30: Complete physicochemical-case and minimum-liver sensitivity results. (Part 2 of 3.)

cohort	within contrast	matched groups
Primary	0.2088	15
Complete size and zeta	0.1465	9
Liver at least 0.01	0.2088	15
Liver at least 0.1	0.2088	15
Liver at least 1	0.2026	14

Table S30: Complete physicochemical-case and minimum-liver sensitivity results. (Part 3 of 3.)

Complete-case restriction can select a different set of experiments because measurement completeness is not randomized. Minimum-liver restrictions select on an observed outcome. Neither restriction should be interpreted as correcting confounding. Their role is to show how cohort definitions affect the descriptive association and how many reference groups continue to support the comparison.

Shuffle summary	Rank correlation
2.5th percentile	0.6253
Median	0.7041
97.5th percentile	0.7726

Table S31: Distribution of rank correlation over 1,000 shuffled-liver diagnostics.

The shuffled-liver procedure preserves the marginal liver distribution and the original tumour numerator. It destroys their observed pairing, including within-publication dependence. A positive correlation can therefore persist solely because the reconstructed ratio contains the tumour term. No p-value against this shuffle distribution is reported: the shuffle does not represent randomized targeting or a biologically exchangeable null population.

T Recalculation and reproducibility contract

The accompanying extension directory separates model scores, row-level predictions, nested learning manifests, descriptor support and permutation results. Project-specific files record calibration bounds and source-time brackets for release, or objective grids, group omissions, numerical balance and paired-organ frontier membership for polymers. These files distinguish evaluated observations from manuscript interpretation.

Contract	Verification
Source cohort and partitions	SHA-256 fixed; original publication roles retained
Primary tree baseline	Five MAEs independently reproduced before extension reporting
Configuration scores	Recomputed from saved held-out endpoint predictions
Nested learning sets	Whole-group nesting and no held-out overlap checked

Table S32: Executable verification of the extended results. (Part 1 of 2.)

Contract	Verification
Project-specific diagnostics	Coverage or paired-organ objectives independently recalculated
Resource budget	Combined measured analysis time and peak process RSS checked

Table S32: Executable verification of the extended results. (Part 2 of 2.)

To reproduce on Windows, run each project's extended.py after the original run.py, then run the scientific and document tests before rebuilding the manuscripts. The extension does not download additional experimental records or modify the original results. File hashes detect a changed cohort, partition manifest or primary score file. Re-execution with a different library version should not silently replace the saved benchmark.

Recalculation establishes consistency of the computational record, not independent experimental replication. The additional tests share source observations with the primary analysis. Their purpose is to expose estimator, representation and objective sensitivity in a form that can inform a separately designed validation study with harmonized measurements.